GENERAL MEDICINE · INTRO

THE BODY RUNS ON ELECTROLYTES.

A 10-minute introduction to membrane potentials, action potentials and the electrolytes that keep cells (and patients!) functioning.

Na^+ K^+ Ca^{2+} Cl^- Mg^{2+}

TODAY'S PLAYLIST · 6 TOPICS

From normal physiology to clinical consequences

01 · Electrical life

Resting potential & action potentials

02 · The voltage duo

Na^+ + K^+

03 · The muscle factor

Calcium

04 · The quiet regulator

Chloride

05 · The heart

Electrical activity in cardiac cells

06 · When balance breaks

The clinical significance

GENERAL MEDICINE · INTRO

The electrical nature of life.

Resting potential & action potentials

Na^+ K^+

01 · THE ELECTRICAL NATURE OF LIFE

The resting membrane potential.

01 · THE ELECTRICAL NATURE OF LIFE

Every cell has an electrical difference.

Resting membrane potential

≈ -70 mV

Typical value for a large myelenated neuron at rest. The inside is negative relative to the outside.

- Unequal ion distribution
- Selective membrane permeability
- Na^+/K^+ pump maintains the gradients

Why is the inside negative?

K^+ leakage

At rest, the membrane is much more permeable to K^+ than to Na^+. K^+ tends to leave the cell, leaving behind relatively more negative charge.

K^+ moves out → inside becomes negative

Na^+-K^+ pump

More positive charges are pumped outside than inside

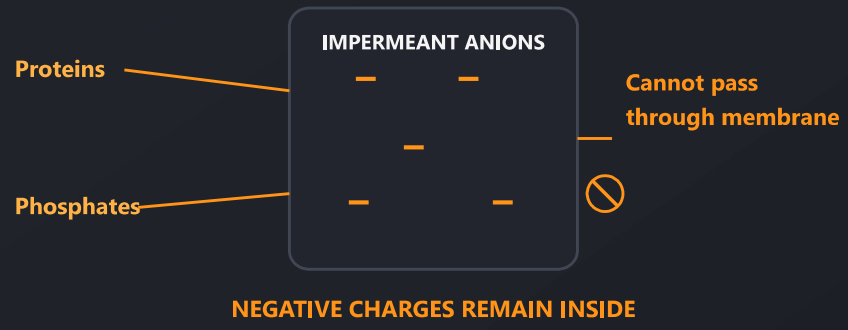


01 · THE ELECTRICAL NATURE OF LIFE

Why is the inside negative?

Impermeant Anions (Gibbs-Donnan Phenomenon)

Inside the cell, are many negatively charged ions that cannot pass through the membrane channels.

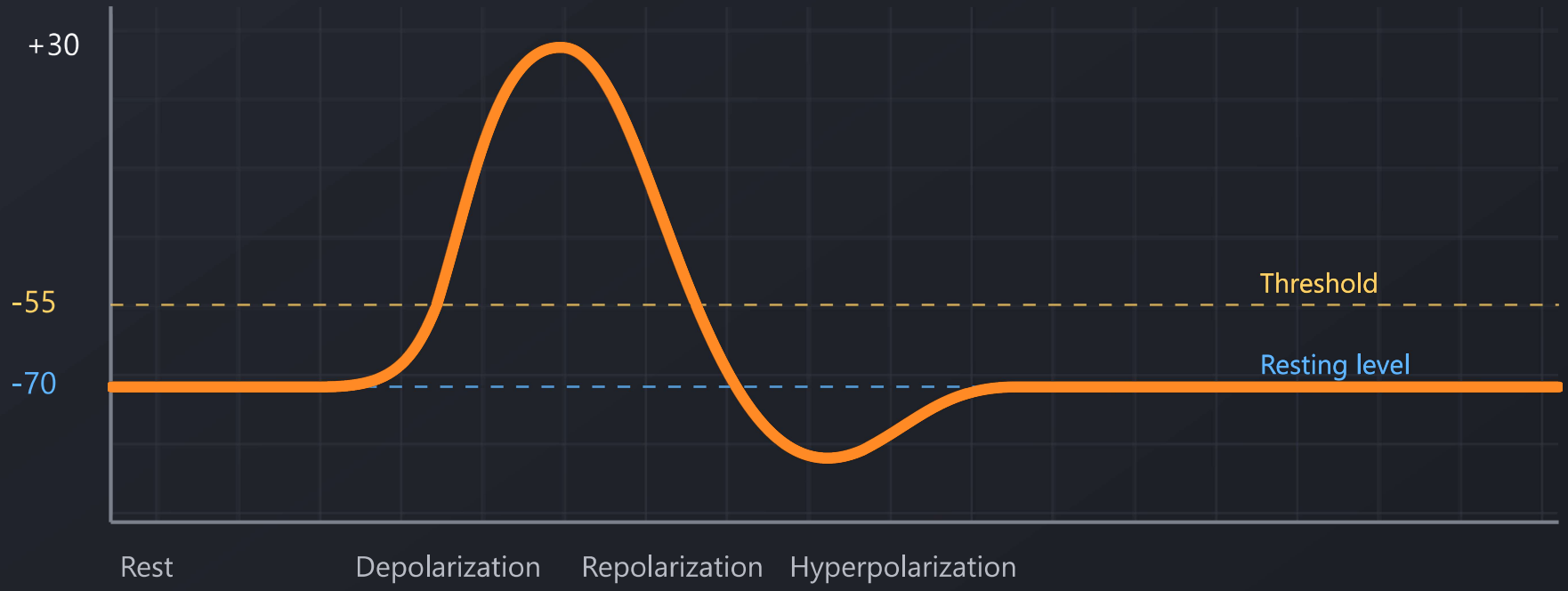


01 · THE ELECTRICAL NATURE OF LIFE

The action potential.

The action potential.

A rapid, temporary change in membrane voltage that carries an electrical signal.



Schematic neuron trace; values and shape are approximate.

01 · THE ELECTRICAL NATURE OF LIFE

The rules of the game.

01 · THE ELECTRICAL NATURE OF LIFE

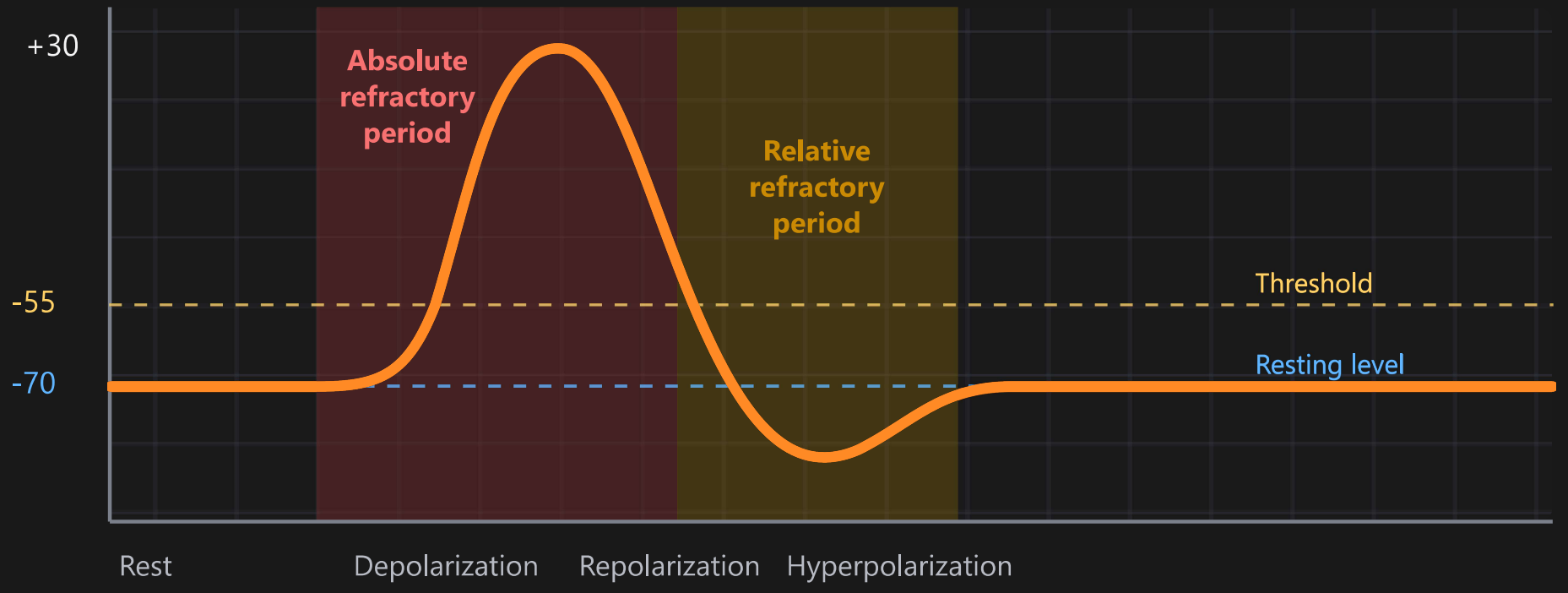
Two rules make the signal reliable.

All-or-none law

If threshold is reached at any point on the membrane of a normal fiber, a full action potential occurs over the entire membrane. If threshold is not reached, it does not. A stronger stimulus does not make that action potential taller; it can increase firing frequency.

Refractory period

The nerve can not respond to a second stimulus during much of the action potential, and even longer. This is referred to as the refactory period.


+30
-55
-70
Absolute refractory period
Relative refractory period
Threshold
Resting level
Rest
Depolarization
Repolarization
Hyperpolarization

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The voltage duo.

$Na^+ + K^+$

Na^+ K^+

02 · THE VOLTAGE DUO

The where of it.

02 · THE VOLTAGE DUO

Every cell has an electrical difference.

Typical concentrations, approximately in mmol/L

Ion	Inside cell (ICF)	Outside cell (ECF)	Driving tendency
Na^+	≈ 10-15	≈ 140-145	Enters when Na^+ channels open
K^+	≈ 140	≈ 4	Leaves through resting leak channels

Sodium

Major ECF cation; dominant determinant of ECF osmolality and important for action-potential upstroke.

Potassium

Major ICF cation; crucial to resting voltage, excitability and repolarization.

02 · THE VOLTAGE DUO

The how of it.

02 · THE VOLTAGE DUO

Na^+/K^+ ATPase

Na^+/K^+ ATPase

- Present on the cell membraness
- Transports 3 Na^+ outside and 2 K^+ inside cell, while breaking down ATP in the process.

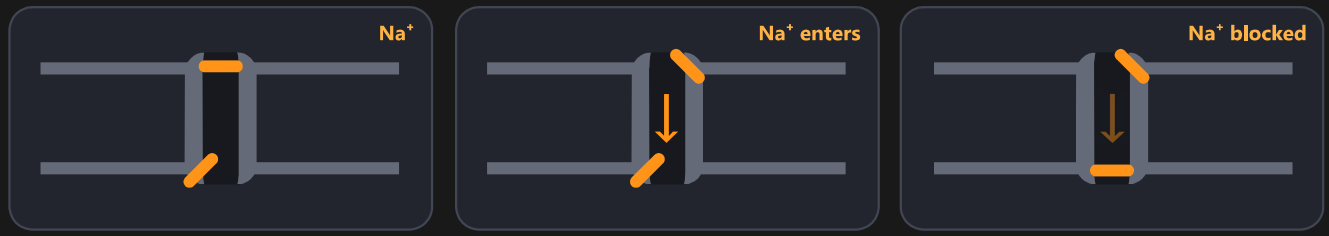
Electrogenic

One net positive charge moved outward per ATP cycle → a small direct negative contribution inside.

Physiological function

Preserves ionic gradients over time

Voltage-gated Na^+ channels

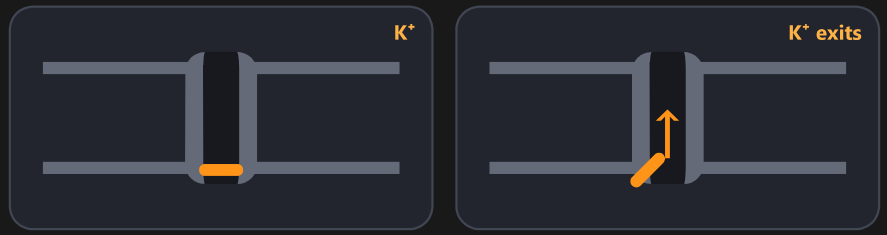


Resting
-90 mV · activation gate closed

Activated
-90 to +35 mV · both gates open

Inactivated
+35 to −90 mV · inactivation gate closed

Voltage-gated K^+ channels



Resting
-90 mV · closed

Slow activation
+35 to -90 mV · open

02 · THE VOLTAGE DUO

The why of it.

02 · THE VOLTAGE DUO

Uses of Sodium and Potassium

Sodium

- Fluid and Osmotic Balance
- Nerve Impulse Transmission & Muscle Contraction
- Cellular Homeostasis and Transport
- Blood Pressure Regulation (Via volume control)
- Renal Acid-Base Balance

Potassium

- Resting Membrane Potential
- Nerve Impulse Transmission & Muscle Contraction
- Blood Pressure Regulation (Via vascular constricton/dialation)
- Cellular Metabolism
- Protein/DNA Synthesis

GENERAL MEDICINE · INTRO

The muscle factor.

Calcium

Ca^{2+}

03 · THE MUSCLE FACTOR

The where of it.

03 · THE MUSCLE FACTOR

Calcium warehouses

Major stores

99% of the body's calcium is stored in:

- Bones
- Teeth

Other stores

The remaining 1% is distributed outside the skeleton:

- Extracellular Fluid
- Intracellular Space
- Plasma
- Cell organelles(mitochondria/endoplasmic reticulum)

03 · THE MUSCLE FACTOR

The why of it.

03 · THE MUSCLE FACTOR

Role of calcium in the body

In Bones and Teeth

Acts as a mechanical framework and a reserve that can be mobilized by hormones like parathyroid hormone (PTH) when blood levels drop.

In Blood and Extracellular Fluid

Mantains vascular tone, nerve impulse conduction, enzyme cofactor activities, and blood coagulation.

In Muscle Cells

Cross-bridge formation and muscle contraction

In Nerves and Synapses

Regulates neurotransmitter release and flows through voltage-gated calcium channels

03 · THE MUSCLE FACTOR

The how of it.

03 · THE MUSCLE FACTOR

Role of calcium in muscle contraction

In Muscle Cells:

Calcium is released from the sarcoplasmic reticulum into the sarcoplasm to bind with troponin, enabling actin-myosin cross-bridge formation and muscle contraction.

Ca^{2+} released from SR → binds troponin → tropomyosin shifts → actin-myosin cross-bridges form → muscle contracts

GENERAL MEDICINE · INTRO

The quiet regulator.

Chloride

Cl^-

04 · THE QUIET REGULATOR

The where of it.

04 · THE QUIET REGULATOR

Chloride warehouses

Major

85% to 90% of the body's chloride is stored in extra cellular fluid:

- Blood plasma
- Interstitial fluid
- Cerebrospinal fluid
- Gastric juices

Other

The remaining 10% to 15% is located inside cells:

- Concentration inside most typical tissue cells is low
- Red blood cells have higher concentrations of chloride in the ICF

04 · THE QUIET REGULATOR

The why of it.

04 · THE QUIET REGULATOR

What does it even do?

Role of chloride in the body:

- Fluid Balance & Osmotic Pressure regulaton
- Acid-Base (pH) Balance regulaton
- Gastric juices production
- Nerve conduction and Muscle contraction
- Carbon Dioxide Transport

04 · THE QUIET REGULATOR

The why now? of it.

04 · THE QUIET REGULATOR

Role of chloride in medcine

Hypochloremia:

- Vomiting
- Volume depletion
- Metabolic alkalosis

Hyperchloremia:

- Bicarbonate loss

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The heart.

Electrical activity in cardiac cells

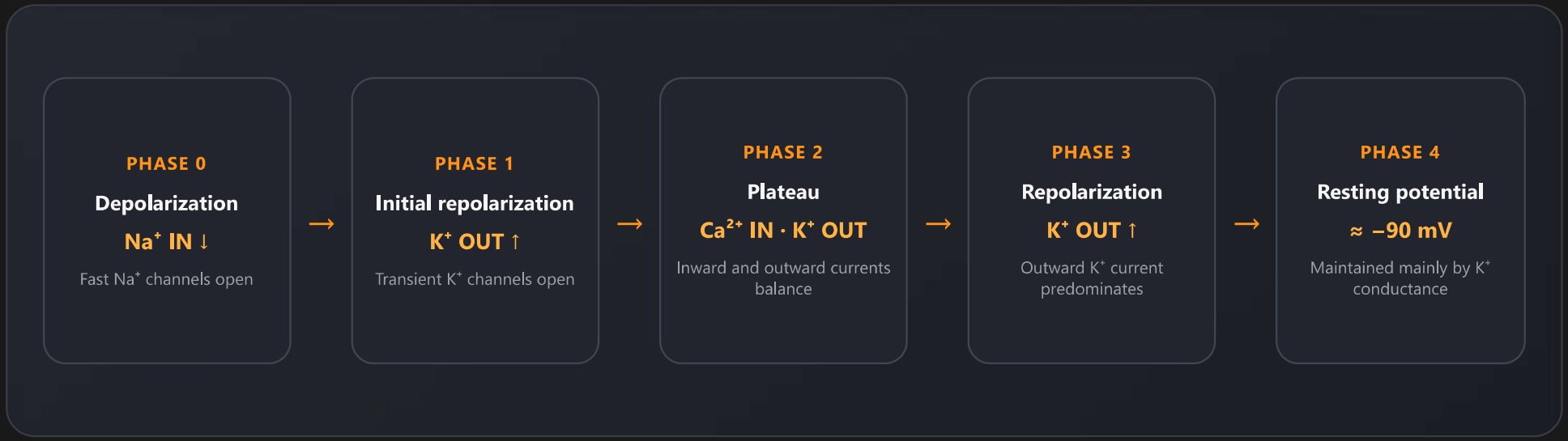
Na^{+} K^{+} Ca^{2+} Cl^{-} Mg^{2+}

05 · THE HEART

The why of it.

05 · THE HEART

Role of electrolytes in the heart



GENERAL MEDICINE · INTRO

When balance breaks.

The clinical significance

Na^{+} K^{+} Ca^{2+} Cl^{-} Mg^{2+}

06 · WHEN BALANCE BREAKS

Sodium

Hyponatremia

Reduced plasma tonicity causes water to enter brain cells, potentially producing cerebral edema and seizures.

Hypernatremia

Increased plasma tonicity draws water out of brain cells, causing cellular dehydration and neurological dysfunction.

Syndrome of Inappropriate ADH(SIADH)

Excessive ADH-mediated water retention produces dilutional hyponatremia.

06 · WHEN BALANCE BREAKS

Potassium

Hypokalemia

Altered membrane excitability and cardiac repolarization can cause muscle weakness, U waves and arrhythmias.

Hyperkalemia

Elevated extracellular K^+ depolarizes the resting membrane potential, potentially impairing cardiac conduction and causing fatal arrhythmias.

Refractory hypokalemia

Concurrent magnesium deficiency promotes renal potassium loss, making hypokalemia difficult to correct.

06 · WHEN BALANCE BREAKS

Calcium

Hypocalcemia

Reduced extracellular ionized calcium increases neuronal excitability, producing tetany and carpopedal spasms.

Hypercalcemia

Elevated calcium decreases neuromuscular excitability, causing muscle weakness, constipation and altered mental status.

QT interval abnormalities

Hypocalcemia prolongs ventricular repolarization and the QT interval, whereas hypercalcemia shortens them.

06 · WHEN BALANCE BREAKS

Chloride

Hypochloremic metabolic alkalosis

Gastric HCl loss during prolonged vomiting causes chloride depletion and promotes renal bicarbonate retention.

Hyperchloremic metabolic acidosis

Increased chloride relative to bicarbonate is associated with normal-anion-gap metabolic acidosis.

Cystic fibrosis

Defective CFTR-mediated chloride transport disrupts epithelial fluid secretion, producing thick mucus and elevated sweat chloride.

06 · WHEN BALANCE BREAKS

Magnesium

Hypomagnesemia

Magnesium deficiency increases neuromuscular excitability and predisposes to cardiac arrhythmias.

Torsades de pointes

Intravenous magnesium sulfate is used to treat this potentially life-threatening polymorphic ventricular tachycardia.

Eclampsia

Magnesium sulfate prevents and treats eclamptic seizures by reducing neuromuscular and central nervous system excitability.